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Israel A. Azime

PhD Candidate | Research Assistant

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EDUCATION

PhD candidate <i>Universität des Saarlandes</i> <i>Thesis: Understanding, Confusion, and Consequence of LLMs in Low Resource Languages</i> <i>Supervisor: Prof. Dr.-Ing. Philipp Slusallek and Prof. Dr. Dietrich Klakow</i>	September 2022 – December 2026[Expected] Saarbrücken, Germany
Master of Science, Mathematical Sciences – Machine Intelligence <i>African Institute for Mathematical Sciences</i> <i>Thesis: Exploring Data Imbalance and Modality Bias in Hateful Memes</i> <i>Supervisor: Prof. Dr. Marcus Rohrbach</i>	October 2019 – March 2021 Accra, Ghana
Bachelor of Science, Software Engineering <i>Addis Ababa University</i> Project: <i>Amharic online handwriting recognition keyboard.</i> Certification Of Recognition For Outstanding Graduation Project <i>Advisor: Prof. Dr. Natnael Argaw Wondimu</i>	September 2012 – December 2017 Addis Ababa, Ethiopia

WORK EXPERIENCE

PhD Internship <i>University of South Florida,</i>	July 2025 – Present <i>Tampa, Florida (remote unpaid from Saarbrücken, Germany)</i>
Socio-Cultural Large Language Models and Low Resource Languages	
Research Assistant HPC cluster support <i>Universität des Saarlandes,</i>	September 2022 – Present Saarbrücken, Germany
- NLP, Multimodal , High performance computing (HPC), AI/ML, Research - Cluster management, Multi GPU model training and HPC related training and support.	
Applied Research Data Scientist <i>iQuartic, New Wave</i>	March 2021 – October 2022 Maryland, USA (Remote)
- Collaborated with a team of data scientists on AI solutions for automated risk adjustment, leading the transition of the data science pipeline from paid to in-house solutions. - Developed and contributed to various projects in medical OCR, medical spell checker, visit recognition, and ICD code extraction. This included working on pipelines for data collection, model building, deployment and testing.	
Applied ML Developer <i>SingularityNet, IcoG-Labs Software Consultancy</i>	June 2017 – August 2022 Addis Ababa, Ethiopia
- Fake news & plant disease detection (97–98% acc.) using CNN, GAN, and VAE models. - Web recommendation via Doc2Vec & DeepCTR; large-scale image retrieval (50 GB). - Holistically-nested edge detection for visual analysis. Graph analytics on Slack data for behavioral & network insights.	

SKILLS

Machine Learning Frameworks	Python, PyTorch, Tensorflow, SciPy, NumPy, Pandas, ApacheSpark, Docker
AI and Research Areas	Deep Learning, Computer Vision, Natural Language Processing, Multimodal Learning
Programming and Tools	Python, Java, C#, Web Dev(HTML, CSS, JS), C/C++ (academic experience)
Soft and Professional Skills	Project Management, Teamwork, Mentoring, Public Speaking, Story Telling, Writing
Languages	Amharic(Native), English(fluent speaker), German (A2)

RESEARCH GRANTS

Google Cloud Credits to support research project titled "African Languages"	USD \$9000	September 2025 - September 2026
AfriFact & AfriGuard: Dataset for Fact-Checking and Robust Refusal	USD \$113,332.50 - Lacuna(Google)	June 2025 - June 2026
PaliGemma Research Grants: in GCP Credits.	USD \$5,000 - Google	Aug 2024 - Aug 2025

PROJECTS AND ACTIVITIES

- **Guest Lecture:** University of South Florida.
- **Peer Review:** ACL ARR (2024,2025,2026), AfricaNLP Workshop (4th, 5th, 6th), and EthioNLP Workshop (August 2025).
- **Community:** Member of AI Grid, MasakhaneNLP, and EthioNLP.
- **Language Coordinator:** Led ML data collection for Lacuna Fund projects.
- **Mentor:** SSA Google Developers ML Bootcamp (Gebeya, 2022).
- **Projects:** Amharic NLP tools, Telegram bots, Unity 3D games, Zindi ML challenges (disaster prediction, DNA classification).
- **Instructor:** Solve-IT National Innovation Competition (2019).
- **Participant:** NASA Space Apps Challenge, Addis Ababa (2018).
- **Scholarships:** PyTorch Facebook Scholarship (Udacity, 2018); Nokia & BIT Scholarship Challenge (Udacity, 2021).

WORKSHOPS ORGANIZED

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| • How to train your own Large Language Model using HPC | Frankfurt, Germany - Dec 2025 |
| • 1st EthioNLP Workshop at ICES22 | Hawassa, Ethiopia - September 2025 |
| • AfricaNLP Workshop ACL 2025 | Vienna, Austria - July 2025 |
| • AfricaNLP Workshop, ICLR 2024 | Vienna Austria - May 2024 |
| • PyTorch for Molecular Property Prediction | Saarland, Germany - Jun 2025 |
| • Fairness & Explainability in LLMs | Saarland, Germany - Oct 2024 |
| • Introduction to In-Context Learning | Saarland, Germany - Jun 2024 |
| • Weights & Biases for Research | Saarland, Germany - Nov 2023 |
| • Introduction to Large Language Models | Saarland, Germany - Sep 2023 |

CERTIFICATIONS

Certification Of Recognition For Outstanding Graduation Project - [↗](#)
Mathematics for machine learning: linear algebra - [↗](#)
Software processes and agile practices - [↗](#)
Introduction to TensorFlow for . . . - [↗](#)
Introduction to game development [↗](#)
Deep Learning Specialization - [↗](#)

Principles of game design - [↗](#)
Html, CSS and Javascript - [↗](#)
R programming - [↗](#)
Linux Fundamentals - [↗](#)
Machine Learning - [↗](#)
Gamification - [↗](#)

CONFERENCE PAPERS

- Azime, I. A., Kanubala, D. D., Afonja, T., Fritz, M., Valera, I., Klakow, D., & Slusallek, P. (2025). Accept or deny? evaluating LLM fairness and performance in loan approval across table-to-text serialization approaches. In C. Christodoulopoulos, T. Chakraborty, C. Rose, & V. Peng (Eds.), *Findings of the association for computational linguistics: Emnlp 2025* (pp. 17478–17503). Association for Computational Linguistics. <https://aclanthology.org/2025.findings-emnlp.947/>
- Azime, I. A., Tonja, A. L., Belay, T. D., Chanie, Y., Balcha, B. F., Abadi, N. H., Ademtew, H. B., Nerea, M. A., Yadeta, D. D., Geremew, D. D., Tesfu, A. A., Slusallek, P., Solorio, T., & Klakow, D. (2025). ProverbEval: Exploring LLM evaluation challenges for low-resource language understanding. In L. Chiruzzo, A. Ritter, & L. Wang (Eds.), *Findings of the association for computational linguistics: Naacl 2025* (pp. 6250–6266). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.findings-naacl.350>
- Belay, T. D., Azime, Israel Abebe, Ayele, A. A., Sidorov, G., Klakow, D., Slusallek, P., Kolesnikova, O., & Yimam, S. M. (2025). Evaluating the capabilities of large language models for multi-label emotion understanding. In O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. D. Eugenio, & S. Schockaert (Eds.), *Proceedings of the 31st international conference on computational linguistics* (pp. 3523–3540). Association for Computational Linguistics. <https://aclanthology.org/2025.coling-main.237/>

- Azime, I. A., Tonja, A. L., Belay, T. D., Fuge, M. Y., Wassie, A. K., Jada, E. S., Chanie, Y., Sewunetie, W. T., & Yimam, S. M. (2024). Walia-LLM: Enhancing Amharic-LLaMA by integrating task-specific and generative datasets. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Findings of the association for computational linguistics: Emnlp 2024* (pp. 432–444). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.findings-emnlp.25>
- Romero, D., Lyu, C., Wibowo, H. A., Lynn, T., Hamed, I., Kishore, A. N., Mandal, A., Dragonetti, A., Abzaliev, A., Tonja, A. L., Balcha, B. F., Whitehouse, C., Salamea, C., Velasco, D. J., Adelani, D. I., Meur, D. L., Villa-Cueva, E., Koto, F., Farooqui, F., ... Aji, A. F. (2024). Cvqa: Culturally-diverse multilingual visual question answering benchmark.
- Tonja, A. L., Azime, I. A., Belay, T. D., Yigezu, M. G., Mehamed, M. A. A., Ayele, A. A., Jibril, E. C., Woldeyohannis, M. M., Kolesnikova, O., Slusallek, P., Klakow, D., & Yimam, S. M. (2024). EthioLLM: Multilingual large language models for Ethiopian languages with task evaluation. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (Irec-coling 2024)* (pp. 6341–6352). ELRA; ICCL. <https://aclanthology.org/2024.i-rec-main.561/>
- Adelani, D. I., Masiak, M., Azime, I. A., Alabi, J., Tonja, A. L., Mwase, C., Ogundepo, O., Dossou, B. F. P., Oladipo, A., Nixdorf, D., Emezue, C. C., Al-azzawi, S., Sibanda, B., David, D., Ndoela, L., Mukiibi, J., Ajayi, T., Moteu, T., Odhiambo, B., ... Stenertorp, P. (2023). MasakhaNEWS: News topic classification for African languages. In J. C. Park, Y. Arase, B. Hu, W. Lu, D. Wijaya, A. Purwarianti, & A. A. Krisnadhi (Eds.), *Proceedings of the 13th international joint conference on natural language processing and the 3rd conference of the asia-pacific chapter of the association for computational linguistics (volume 1: Long papers)* (pp. 144–159). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.ijcnlp-main.10>
- Tonja, A. L., Belay, T. D., Azime, I. A., Ayele, A. A., Mehamed, M. A., Kolesnikova, O., & Yimam, S. M. (2023). Natural language processing in Ethiopian languages: Current state, challenges, and opportunities. In R. Mabuya, D. Mthobela, M. Setaka, & M. Van Zaanen (Eds.), *Proceedings of the fourth workshop on resources for african indigenous languages (rail 2023)* (pp. 126–139). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.rail-1.14>

WORKSHOP PAPERS

- Azime, I. A., Angamo, A. K., Tamiru, H. M., Marilign, D. M., Slusallek, P., Yimam, S. M., & Klakow, D. (2026). AmharicStoryQA: A Multicultural Story Question Answering Benchmark in Amharic. *arXiv preprint arXiv:2602.02774*. <https://arxiv.org/abs/2602.02774>
- Azime, I. A., Belay, T. D., & Tonja, A. L. (2025). Evaluation sheet for deep research: A use case for academic survey writing. In C. Zhang, E. Allaway, H. Shen, L. Miculicich, Y. Li, M. M'hamdi, P. Limkonchotiwat, R. H. Bai, S. T.y.s.s., S. S. Han, S. Thapa, & W. B. Rim (Eds.), *Proceedings of the 9th widening nlp workshop* (pp. 242–250). Association for Computational Linguistics. <https://aclanthology.org/2025.winlp-main.36/>
- Azime, I. A., Al-azzawi, S., Tonja, A. L., Shode, I., Alabi, J., Awokoya, A., Oduwole, M., Adewumi, T., Fanijo, S., & Oyinkansola, A. (2023). Masakhane-afrisenti at SemEval-2023 task 12: Sentiment analysis using Afro-centric language models and adapters for low-resource African languages. In A. K. Ojha, A. S. Doğruöz, G. Da San Martino, H. Tayyar Madabushi, R. Kumar, & E. Sartori (Eds.), *Proceedings of the 17th international workshop on semantic evaluation (semeval-2023)* (pp. 1311–1316). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.semeval-1.182>
- Azime, I. A., & Mohammed, N. (2021). An amharic news text classification dataset. *arXiv preprint arXiv:2103.05639*.

PREPRINT

- Abdullah, B. M., Azime, I. A., Tonja, A. L., Alabi, J. O., Alemu, A. M., Hagos, E. G., Balcha, B. F., Nerea, M. A., Yadeta, D. D., Marilign, D. M., et al. (2026). Ethio-asr: Joint multilingual speech recognition and language identification for ethiopian languages. *arXiv preprint arXiv:2603.23654*. <https://arxiv.org/abs/2603.23654>
- Azime, I. A., Alabi, J. O., Zhang, C., Maab, I., Tonja, A. L., Belay, T. D., Alabi, F. P., Osei, S., Aliyu, S. M., Aguobi, N. F., et al. (2026). Afrifact: Cultural information retrieval, evidence extraction and fact checking for african languages. *arXiv preprint arXiv:2604.00706*.
- Azime, I. A., Belay, T. D., Klakow, D., Slusallek, P., & Chhabra, A. (2025). Bridging the culture gap: A framework for llm-driven socio-cultural localization of math word problems in low-resource languages. *arXiv preprint arXiv:2508.14913*.

REFERENCES

Available upon Request